

2006



Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen
- Write your Student Number on every page
- All questions may be attempted.
- Begin each question in a new booklet.
- All necessary working must be shown.
- Marks may be deducted for careless or poorly presented work.
- Board-approved calculators may be used.
- A list of standard integrals is included at the end of this paper.
- The mark allocated for each question is listed at the side of the question.

Students are advised that this is a Trial Examination only and does not necessarily reflect the content or format of the Higher School Certificate Examination.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Question 1 – (15 marks) – Start a new booklet

Marks

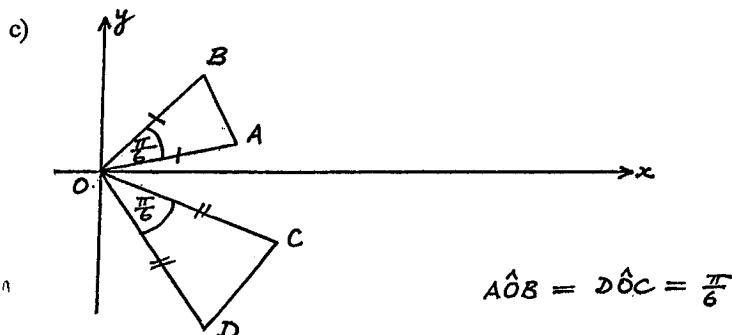
a) Prove that for complex numbers z , z_1 and z_2

(i) $z\bar{z} = |z|^2$ (ii) $\overline{z_1 z_2} = \overline{z_1} \overline{z_2}$

1, 1

b) Use the above two results to prove that $|z_1 z_2| = |z_1| |z_2|$

2



The points A, B, C, D and O are points in the Argand plane such that $OD = OC$ and $OA = OB$. A corresponds to the complex number z and C corresponds to the complex number q . Let $w = \cos \frac{\pi}{6} + i \sin \frac{\pi}{6}$.

(i) Explain why B corresponds to zw .

1

(ii) Find the complex number corresponding to D .

1

(iii) Prove, using complex numbers, that $BC = AD$.

1

d) (i) If α is a double zero of a polynomial $P(x)$ show that α is a single zero of $P'(x)$

2

(ii) Find the integers 'a' and 'b' given that $(x+1)^2$ is a factor of $x^5 + 2x^2 + ax + b$

3

e) (i) Show that $z = 1 + i$ is a root of the equation $z^2 - (3 - 2i)z + (5 - i) = 0$

3

(ii) Find the other root of the equation.

Question 2 – (15 marks) – Start a new booklet

Marks

a) (i) Express $\frac{3x+7}{(x+1)(x+2)(x+3)}$ in partial fractions (show all working)

3

(ii) Hence evaluate $\int_0^1 \frac{3x+7}{(x+1)(x+2)(x+3)} dx$

b) (i) Simplify $\sin(A+B) + \sin(A-B)$

3

(ii) Hence evaluate $\int_0^{\frac{\pi}{4}} \sin 5x \cos 3x dx$

c) Consider the ellipse $E: \frac{x^2}{9-\lambda} + \frac{y^2}{\lambda-4} = 1$ where λ is a real number.

(i) Find the range of possible values of λ

1

(ii) For this range of values of λ , find the volume $V(\lambda)$ in simplest factored form, when the area enclosed by this ellipse is rotated about the x -axis.

3

(iii) Sketch the ellipse E when $\lambda = 5$ clearly showing intercepts, foci and directrices.

5

Question 3 – (15 marks) – Start a new booklet

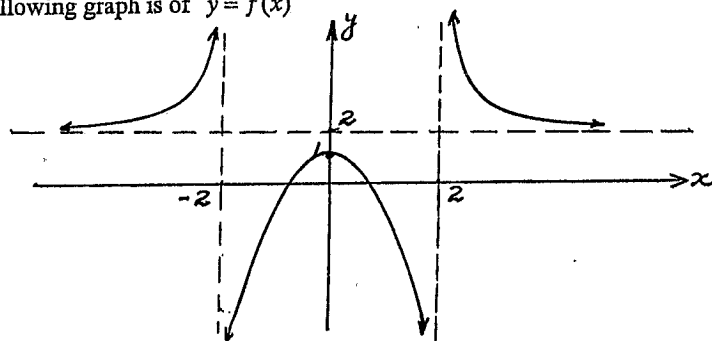
Marks

- a) (i) Sketch $y = [\sin^{-1} x]^{-1}$ clearly demonstrating its relationship to $y = \sin^{-1} x$ 2
- (ii) Sketch $y^2 = \tan^{-1} x$ clearly showing its relationship to $y = \tan^{-1} x$ 2
- b) $P(x)$ is an even monic polynomial of degree four with integer co-efficients. One zero is $3i$ and the product of the zeros is -18 . Factor $P(x)$ fully over the real field. 3

- c) The base of a solid is the circle $x^2 + y^2 = 16$ 4

Find the volume of the solid if every cross-section perpendicular to the x -axis is an equilateral triangle with one side in the base of the solid.

- d) The following graph is of $y = f(x)$



- (i) Suggest a possible expression for $f(x)$ giving a clear explanation of your choice. 2
- (ii) Sketch $y = f'(x)$ 2

Question 4 – (15 marks) – Start a new booklet

Marks

- a) If α, β and γ are the roots of $x^3 - 2x^2 + 4x + 2 = 0$ find the polynomial equation with roots 4

(i) $\alpha - 1, \beta - 1, \gamma - 1$

(ii) $\alpha^2, \beta^2, \gamma^2$

- b) Evaluate

(i) $\int_2^3 \frac{x^3}{x^2 - 1} dx$ 2

(ii) $\int_0^2 x e^x dx$ 2

- c) Consider the curve $C: x^2 + xy + y^2 = 3$

(i) Find $\frac{dy}{dx}$ 1

- (ii) Find all stationary points and points where $\frac{dy}{dx}$ is not defined. 4

- (iii) Sketch C clearly showing the above features and intercepts on the x, y axes. 2

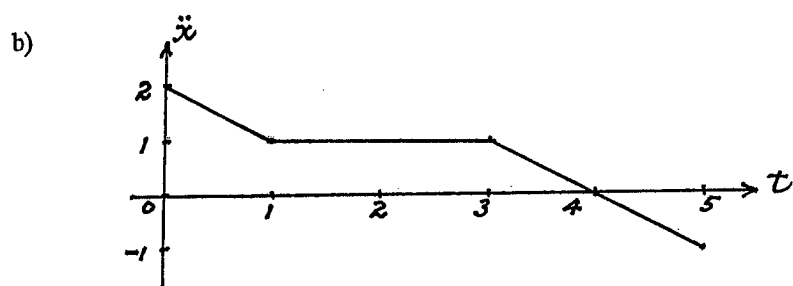
Question 5 – (15 marks) – Start a new booklet

Marks

a) (i) If $I_n = \int_0^{\frac{\pi}{2}} x^n \sin x \, dx$, $n \geq 0$ prove that for $n \geq 2$, $I_n + n(n-1)I_{n-2} = n\left(\frac{\pi}{2}\right)^{n-1}$ 5

(ii) Calculate $\int_0^{\frac{\pi}{2}} x \sin x \, dx$

(iii) Deduce that $\int_0^{\frac{\pi}{2}} x^3 \sin x \, dx = \frac{3}{4}\pi^2 - 6$



A particle starts from rest at the origin. The above sketch represents its acceleration $\ddot{x}$ as a function of time.

(i) Carefully describe the velocity of the particle for $1 \leq t \leq 3$. 2

(ii) Find the velocity of the particle at $t = 5$ 2

c) The region, in the first quadrant, bounded by the curve $y = \cos^{-1} x$ and the co-ordinate axes is rotated about the line $x = -1$. Use the method of cylindrical shells, clearly showing all working, to find the volume of the solid generated. 6

Question 6 – (15 marks) – Start a new booklet

Marks

a) A solid is formed by rotating the ellipse $E: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ about the line $x = 2a$. 4
Find the volume of this solid by taking slices perpendicular to the axis of rotation.

b) The point $P(x_1, y_1)$ lies on the hyperbola $H: \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. ☐

The foci of H are S and S' .

The tangent to H at P cuts the x -axis at T .

(i) Prove that the equation of the tangent at P is $\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$ 2

(ii) Find the coordinates of T . 1

(iii) Prove that $\frac{PS}{PS^1} = \frac{TS}{TS^1}$ 3

c) A particle of mass m kg is dropped from rest in a medium where the resistance force is mkv . ☐

(i) Find the equation of motion and the terminal velocity v_T 2

(ii) Find the time taken for the particle to reach a velocity of $\frac{v_T}{2}$. 3

Question 7 – (15 marks) – Start a new booklet

Marks

- a) A mass m kg is projected vertically upwards with velocity V m/s in a medium with resistance force mkv . Find the maximum height reached. 4
- b) In any triangle ABC prove that $\tan A + \tan B + \tan C = \tan A \tan B \tan C$ 3
- c) (i) If $t = \tan x$ prove that $\tan 4x = \frac{4t(1-t^2)}{t^4 - 6t^2 + 1}$ 2
- (ii) If $\tan x \tan 4x = 1$ deduce that $5t^4 - 10t^2 + 1 = 0$ 1
- (iii) Use basic trigonometry to prove that both $x = 18^\circ$ and $x = 54^\circ$ satisfy the equation $\tan x \tan 4x = 1$ 2
- (iv) Deduce that $\tan 54^\circ = \sqrt{\frac{5+2\sqrt{5}}{5}}$ 3

Question 8 – (15 marks) – Start a new booklet

Marks

- a) (i) Find the sum $1 + 10 + 10^2 + \dots + 10^n$ 7
- (ii) Use the method of mathematical induction to show that
- $$1 \times 9^2 + 11 \times 9^2 + 111 \times 9^2 + \dots + \underbrace{111\dots 1}_{n \text{ ones}} \times 9^2$$
- $$= 10^{n+1} - 9n - 10 \text{ for all positive integers } n \geq 1$$
- b) It is given that $z^5 = 1$ where $z \neq 1$
- (i) Show that $z^2 + z + 1 + z^{-1} + z^{-2} = 0$ 1
- (ii) Show that $z + z^{-1} = 2 \cos \frac{2k\pi}{5}$ $k = 1, 2, 3, 4$ 2
- (iii) By letting $x = z + z^{-1}$ reduce the equation in (i) above to a quadratic equation in x . 3
- (iv) Hence deduce that $\cos \frac{\pi}{5} \cdot \cos \frac{2\pi}{5} = \frac{1}{4}$ 2

Ext. 2 Maths

Q1

(i) $z \bar{z} = (x+iy)(x-iy)$

$$= (x^2 + y^2)^2$$

$$(ii) \overline{z_1 z_2} = (x_1 + iy_1)(x_2 + iy_2)$$

$$= x_1 x_2 - y_1 y_2 + i(x_2 y_1 + x_1 y_2)$$

$$= x_1 x_2 - y_1 y_2 - i(x_2 y_1 + x_1 y_2) \quad \neq$$

and $\bar{z}_1 \bar{z}_2 = (\overline{x_1 + iy_1})(\overline{x_2 + iy_2})$

$$= (x_1 - iy_1)(x_2 - iy_2)$$

$$= x_1 x_2 - i x_1 y_2 - i y_1 x_2 - y_1 y_2$$

$$= x_1 x_2 - y_1 y_2 - i(x_2 y_1 + x_1 y_2)$$

$$= \sqrt{z_1 z_2} \quad \text{from } *$$

b) $z_1 z_2 \overline{z_1 z_2} = |z_1 z_2|^2$ from a)(i)

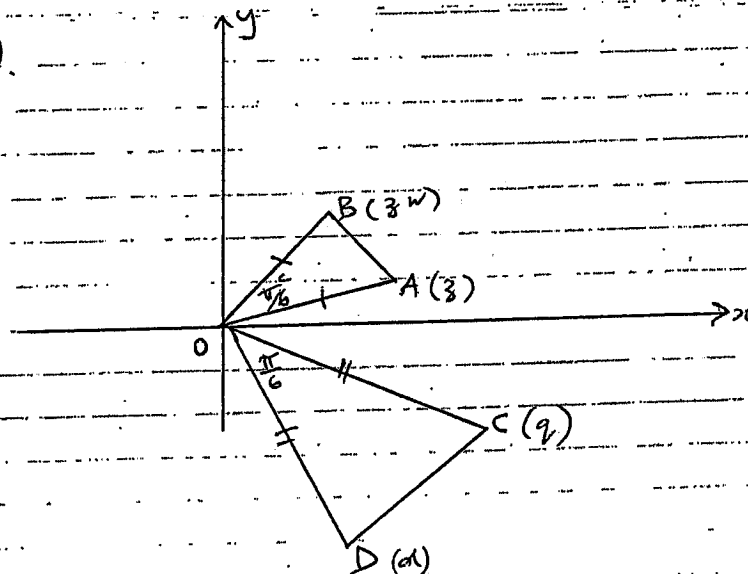
Now, $z_1 z_2 \overline{z_1 z_2} = z_1 z_2 \overline{z_1} \overline{z_2}$ from 2)(ii)

$$= \{z_1, \bar{z}_1\} \{z_2, \bar{z}_2\} \quad \text{Commutative law of } \times \text{ of complex n}$$

$$|z_1 z_2|^2 = |z_1|^2 |z_2|^2$$

$13, 2, 1 = 13, 1/3, 1$ on taking sq. roots.

c) (i).



(i) B represents $3 \times 1 \text{ cis } \frac{\pi}{6}$ since to obtain the

pos'n of B we have rotated A through $\frac{\pi}{6}$

$$[r \cos \theta \times \cos \phi = r \cos (\theta + \phi)]$$

∴ B corresponds to 3ω .. since $\omega = \text{dist } \frac{\pi}{6}$

(ii). Let d be the complex number corresponding to point D.

$$\therefore z \times \text{cis } \frac{\pi}{6} = 2$$

$$\therefore d = \frac{q}{\cos \frac{\pi}{6}} = \frac{q}{W}$$

$$(iii) \quad \vec{BC} + \vec{OB} = \vec{OC}$$

$$\therefore \vec{BC} = \vec{OC} - \vec{OB}$$

$$= \frac{1}{2} - \frac{1}{2}w$$

$$\therefore BC = |2 - 3w|$$

$$\vec{OA} + \vec{AD} = \vec{OD}$$

$$\vec{AD} = \vec{OD} - \vec{OA}$$

$$= d - z.$$

$$= \frac{9}{14} - \frac{1}{2}$$

$$= \frac{2-3W}{W}$$

$$AD = 12.3 \text{ W}$$

$$= |9-3w| \quad \text{since } |w| = |e^{i5\pi/6}|$$

$$= \sqrt{\cos^2 \frac{\pi}{6} + \sin^2 \frac{\pi}{6}}$$

$$= \sqrt{1}$$

$$= 1.$$

$$\therefore BC = AD$$

d) (i) $P(x) = (x-\alpha)^2 Q(x)$, where $Q(\alpha) \neq 0$.

$$\therefore P'(x) = 2(x-\alpha)Q(x) + (x-\alpha)^2 Q'(x)$$

$$= (x-\alpha)[2Q(x) + (x-\alpha)Q'(x)]$$

Now, $Q(\alpha) \neq 0$ $\therefore$ there is no factor of $x-\alpha$ in $2Q(x) + (x-\alpha)Q'(x)$.

$\therefore P'(x)$ only has a single zero of $x=\alpha$

(ii) $(x+1)^2$ is a factor of $P(x) = x^5 + 2x^2 + ax + 1$

$x+1$ is a factor of $P'(x)$ for (i)

$\therefore P'(-1) = 0$ (factor theorem)

Now, $P'(x) = 5x^4 + 4x + a$

$$\therefore P'(-1) = 5 - 4 + a = 0$$

$$\therefore a = -1$$

$$\therefore P(x) = x^5 + 2x^2 - x + b = (x+1)^2 Q(x)$$

$$P(-1) = 0$$

$$(-1)^5 + 2(-1)^2 - (-1) + b = 0$$

$$-1 + 2 + 1 + b = 0$$

$$\therefore b = -2$$

e) (i) Let $P(z) = z^2 - (3-2i)z + (5-i) = 0$

$$\therefore P(1+i) = (1+i)^2 - (3-2i)(1+i) + 5-i$$

$$= 1+2i-1 - 3-3i+2i-2+5-i$$

$$= 0$$

$z = 1+i$ is a root of $z^2 - (3-2i)z + 5-i$

(ii) Let α be the other root
 $\therefore \alpha + 1+i = 3-2i$ ($= -\frac{b}{a}$)
 $\therefore \alpha = 2-3i$

(OR divide $P(z)$ by $z-1-i$)

Q2
 a) (i) $\frac{3x+7}{(x+1)(x+2)(x+3)} = \frac{a}{x+1} + \frac{b}{x+2} + \frac{c}{x+3}$

$3x+7 = a(x+2)(x+3) + b(x+1)(x+3) + c(x+1)(x+2)$

Let $x = -2$

$1 = b(-1)(1) \therefore b = -1$

$x = -1$

$4 = a(1)(2) \therefore a = 2$

$x = -3$

$-2 = c(-2)(-1) \therefore c = -1$

$\frac{3x+7}{(x+1)(x+2)(x+3)} = \frac{2}{x+1} - \frac{1}{x+2} - \frac{1}{x+3}$

(ii) $\int_0^1 \frac{3x+7}{(x+1)(x+2)(x+3)} dx = \int_0^1 \left[\frac{2}{x+1} - \frac{1}{x+2} - \frac{1}{x+3} \right] dx$
 $= [2 \ln(x+1) - \ln(x+2) - \ln(x+3)]_0^1$

$= 2 \ln 2 - \ln 3 - \ln 4 - 2 \ln 1 + \ln 2 + \ln 3$
 $= 3 \ln 2 - 2 \ln 2$
 $= \ln 2$

b) (i) $\sin(A+B) + \sin(A-B)$
 $= \sin A \cos B + \cos A \sin B + \sin A \cos B - \cos A \sin B$
 $= 2 \sin A \cos B$

(ii) $\sin A \cos B = \frac{1}{2} (\sin(A+B) + \sin(A-B))$

$\int_0^{\frac{\pi}{4}} \sin 5x \cos 3x dx = \frac{1}{2} \int_0^{\frac{\pi}{4}} (\sin 8x + \sin 2x) dx$

($A = 5x, B = 3x$)

$= \frac{1}{2} \left[-\frac{1}{8} \cos 8x - \frac{1}{2} \cos 2x \right]_0^{\frac{\pi}{4}}$

2

$= \frac{1}{2} \left(-\frac{1}{8} \cos 2\pi - \frac{1}{2} \cos \frac{\pi}{2} + \frac{1}{8} + \frac{1}{2} \right)$

$= \frac{1}{2} \left(-\frac{1}{8} - 0 + \frac{1}{8} + \frac{1}{2} \right)$

$= \frac{1}{4}$

c) (i) We require $9-\lambda > 0 \wedge \lambda-4 > 0$

$\underline{0 < \lambda < 9} \wedge \underline{\lambda > 4}$

$\underline{4 < \lambda < 9}$

(ii) $V = 2\pi \int_0^{\sqrt{9-\lambda}} y^2 dx$ $\frac{x^2}{2-\lambda} + \frac{y^2}{\lambda-4} = 1$

$= 2\pi \int_0^{\sqrt{9-\lambda}} (\lambda-4)(\lambda-4) \left[1 - \frac{x^2}{9-\lambda} \right] dx$

$= 2\pi \left[(\lambda-4)x - \frac{\lambda-4}{9-\lambda} \frac{x^3}{3} \right]_0^{\sqrt{9-\lambda}}$

3 $= 2\pi \left[(\lambda-4)\sqrt{9-\lambda} - \frac{\lambda-4}{9-\lambda} \frac{(9-\lambda)\sqrt{9-\lambda}}{3} \right]$

$= 2\pi (\lambda-4) \left[\frac{2}{3} \sqrt{9-\lambda} \right]$

$= \frac{4\pi}{3} (\lambda-4) \sqrt{9-\lambda} \text{ units}^3$

(iii) If $\lambda=5$, E becomes

$$\frac{x^2}{4} + \frac{y^2}{1} = 1$$

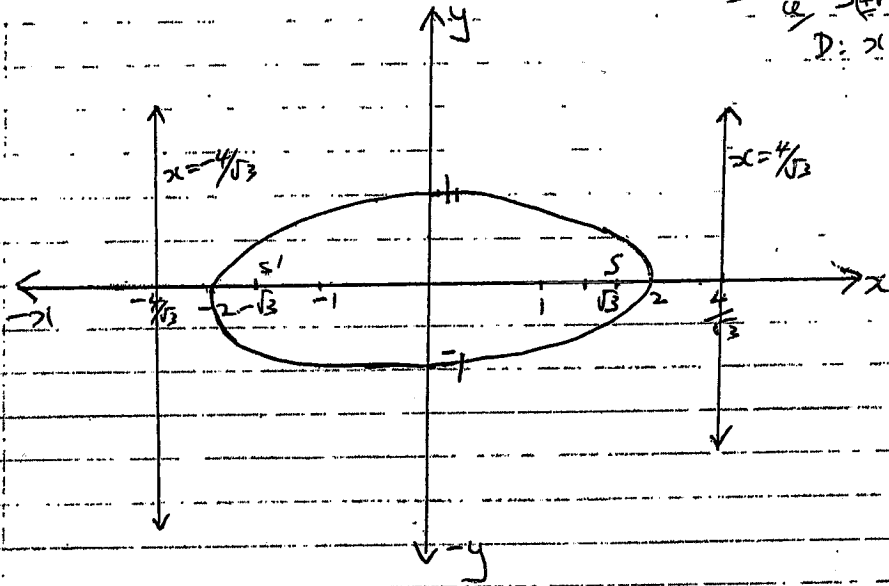
i.e. $a=2, b=1$
 Now, $b^2 = a^2(1-e^2)$
 $\therefore 1 = 4(1-e^2)$
 $\frac{1}{4} = 1-e^2$

$$e^2 = \frac{3}{4}$$

$$e = \frac{\sqrt{3}}{2}$$

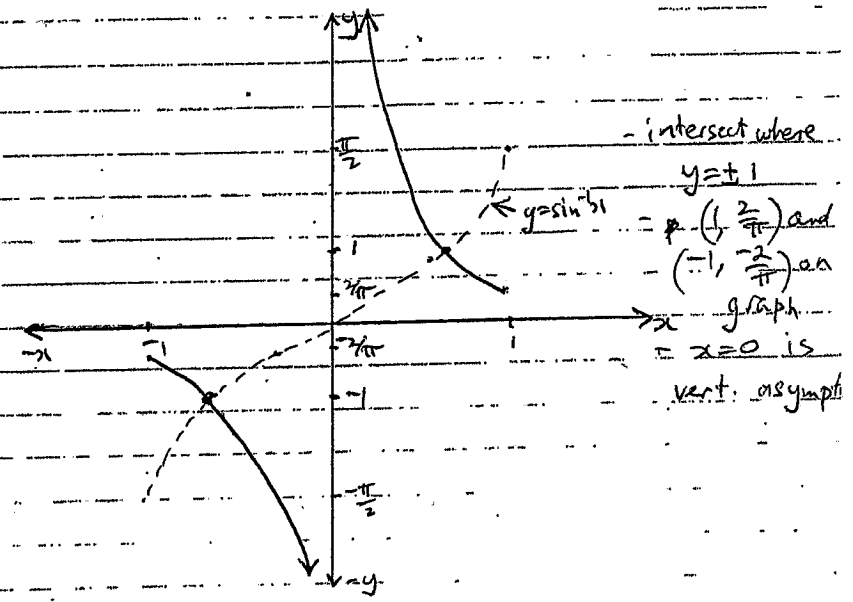
$$\therefore ae = \sqrt{3}, \quad \frac{a}{e} = \frac{4}{\sqrt{3}}$$

i.e. $S(\pm\sqrt{3}, 0)$
 $D: x = \pm \frac{4}{\sqrt{3}}$



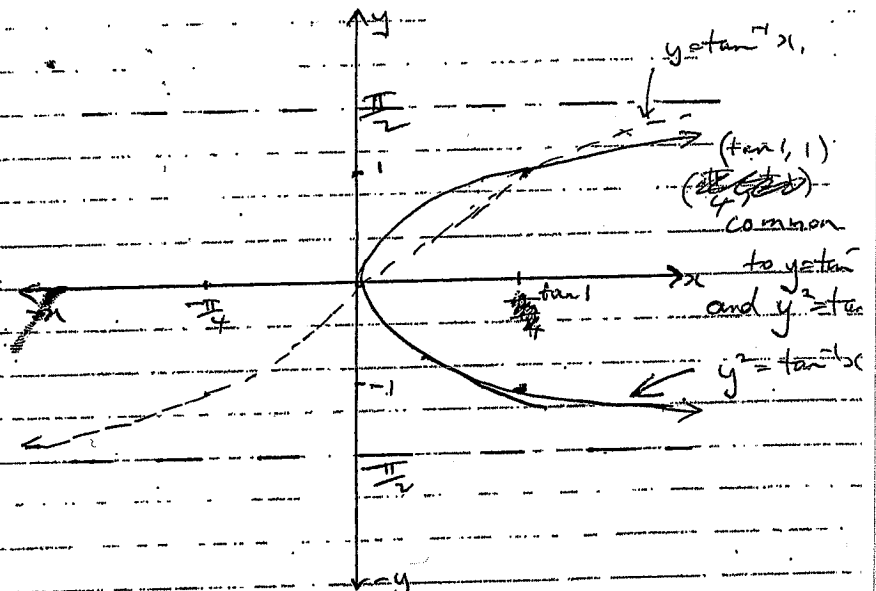
Q3
a) (i)

2



(ii)

2



b) Let $P(x) = x^4 + ax^2 + b$ a, b integers.
 If $x = 3i$ is a zero, so is $-3i$
 (conjugate root theorem).

$(x+3i)(x-3i)$ is a factor
 i.e. $x^2 + 9$

Let the other 2 roots be α, β
 $\therefore 3i \cdot x - 3i \cdot x \cdot \alpha \beta = b = -18$
 $\therefore 9\alpha\beta = -18$
 $\alpha\beta = -2$ ①

Also, $\alpha + \beta + 3i - 3i = 0$
 $\therefore \alpha + \beta = 0$ ②

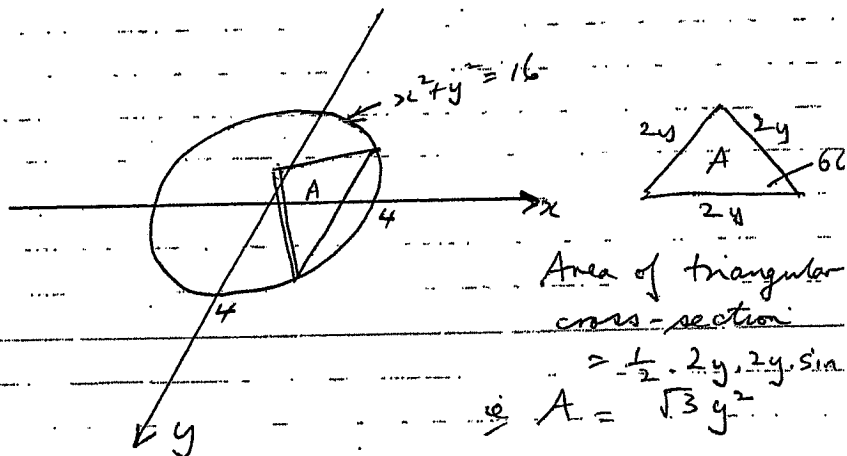
Solve ①, ② simultaneously
 $\Rightarrow \alpha(-\alpha) = -2$
 $\alpha^2 = 2$
 i.e. $\alpha = \sqrt{2}, \beta = -\sqrt{2}$

$$P(x) = (x+3i)(x-3i)(x+\sqrt{2})(x-\sqrt{2})$$

$$= (x^2 + 9)(x^2 - 2)$$

$$= x^4 + 7x^2 - 18$$

c)



$$\Delta V = A \Delta x$$

$$= \sqrt{3}y^2 \Delta x$$

$$= \sqrt{3}(16-x^2) \Delta x$$

$$\therefore V = \lim_{\Delta x \rightarrow 0} \sqrt{3} \sum_{x=-4}^4 (16-x^2) \Delta x$$

$$= \sqrt{3} \int_{-4}^4 (16-x^2) dx$$

$$= 2\sqrt{3} \int_0^4 (16-x^2) dx$$

$$= 2\sqrt{3} \left[16x - \frac{x^3}{3} \right]_0^4$$

$$= 2\sqrt{3} \left(64 - \frac{64}{3} \right)$$

$$= 2\sqrt{3} \times \frac{2}{3} \times 64$$

$$= \frac{256\sqrt{3}}{3} \text{ u}^3$$

d) (i) Horizontal asymptote $y=2$
 i.e. $y \rightarrow 2$ as $x \rightarrow \pm \infty$
 Vertical asymptotes $x = \pm 2$
 i.e. denominator contains $x^2 - 4$
 Passes through $(0, 1)$.

$$y = \frac{2x^2 + a}{x^2 - 4}$$

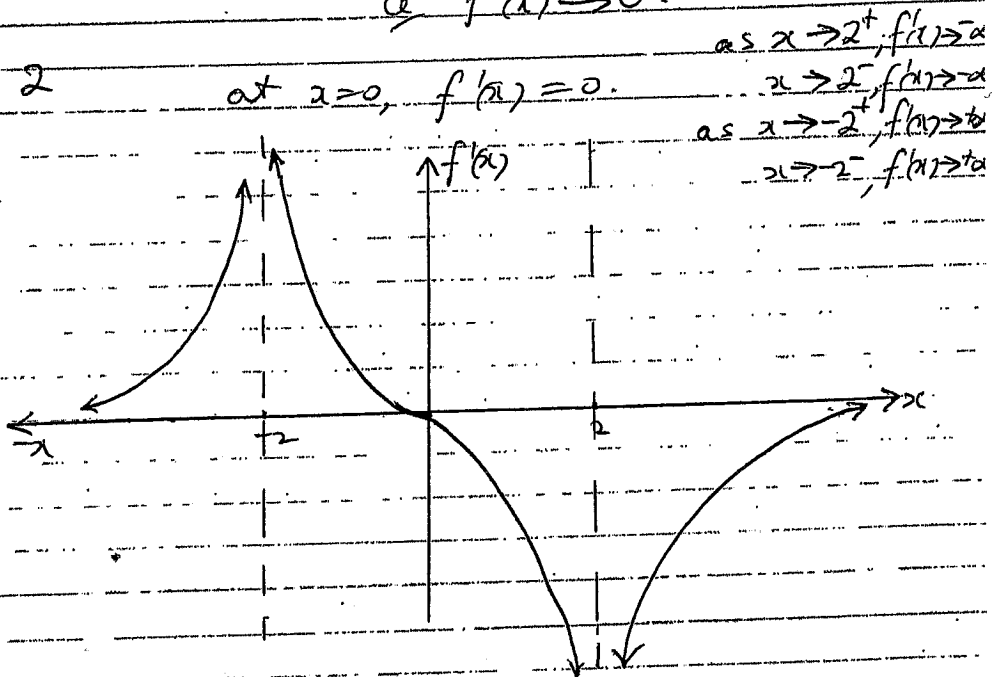
2. when $x=0, y=1$
 $1 = \frac{a}{-4}$
 $a = -4$
 $\therefore y = \frac{2x^2 - 4}{x^2 - 4}$

(ii) $f(x)$ decreasing for $x > 0$
 " increasing " $x < 0$

$$\therefore f'(x) > 0 \text{ for } x < 0$$

$$f'(x) < 0 \text{ for } x > 0$$

as $x \rightarrow \pm\infty$, $f(x) \rightarrow \text{horizontal line}$
 i.e. $f'(x) \rightarrow 0$.



Q4
 a) (i) $P(x) = x^3 - 2x^2 + 4x + 2 = 0$ has roots α, β, γ .
 We want eqn in x where $x = \alpha - 1$.

Now, $P(\alpha) = 0$

$$\therefore P(x+1) = 0$$

$$P(x+1) = (x+1)^3 - 2(x+1)^2 + 4(x+1) + 2$$

$$= x^3 + 3x^2 + 3x + 1 - 2x^2 - 4x - 2 + 4x + 6$$

(2)

i.e. required eqn is

$$x^3 + x^2 + 3x + 5 = 0$$

(ii) $x = \alpha^2$

$$\alpha = \pm\sqrt{x}$$

$$P(\alpha) = 0 \therefore P(\pm\sqrt{x}) = 0$$

$$\text{i.e. } (\pm\sqrt{x})^3 - 2(\pm\sqrt{x})^2 + 4(\pm\sqrt{x}) + 2 = 0$$

$$\text{i.e. } \pm\sqrt{x}(x+4) = 2x-2$$

$$\therefore x(x+4)^2 = (2x-2)^2$$

$$x(x^2 + 8x + 16) = 4x^2 - 8x + 4$$

(2)

$$x^3 + 8x^2 + 16x = 4x^2 - 8x + 4$$

$$\text{i.e. } x^3 + 4x^2 + 24x - 4 = 0$$

b) (i) $\int_2^3 \frac{x^3}{x^2-1} dx = \int_2^3 \frac{x(x^2-1) + x}{x^2-1} dx$

$$= \int_2^3 x + \frac{x}{x^2-1} dx$$

$$= \left[\frac{1}{2}x^2 + \frac{1}{2}\ln(x^2-1) \right]_2^3$$

(2)

$$= \frac{9}{2} + \frac{1}{2}\ln 8 - 2 - \frac{1}{2}\ln 3$$

$$= 2\frac{1}{2} + \frac{1}{2}\ln \frac{8}{3}$$

$$= 5 + \ln \left(\frac{8}{3} \right)$$

Q.4
b)

$$(ii) \int_0^2 x e^x dx$$

$$= \int_0^2 x \frac{d}{dx} e^x dx$$

$$= x e^x \Big|_0^2 - \int_0^2 e^x \cdot 1 dx$$

$$= [x e^x - e^x]_0^2$$

(2)

$$= 2e^2 - e^2 + 1$$

$$= e + 1$$

Q. c) (i) $x^2 + xy + y^2 = 3$ *

$$\frac{d}{dx} (x^2 + xy + y^2) = \frac{d}{dx} 3$$

$$\text{ie } 2x + x y' + y + 2y y' = 0$$

$$(1) \quad y' (x + 2y) = -2x - y$$

$$\therefore y' = \frac{-2x - y}{x + 2y}$$

$$(ii) \quad y' = 0 \Rightarrow -2x - y = 0$$

$$\text{ie } y = -2x \quad \oplus$$

$$\text{Sub. } y = -2x \text{ in } *$$

$$\Rightarrow x^2 + x(-2x) + 4x^2 = 3$$

$$3x^2 = 3$$

$$x = \pm 1$$

$$\text{when } x = 1, y = -2$$

$$x = -1, y = 2$$

(2)

$\therefore (1, -2)$ & $(-1, 2)$ are st. pts.

Also, $\frac{dy}{dx}$ undefined where $x + 2y = 0$

$$\text{ie } x = -2y$$

Sub. into *

$$\Rightarrow 4y^2 - 2y^2 + y^2 = 3$$

$$3y^2 = 3$$

$$y = \pm 1$$

$$\text{when } y = 1, x = -2$$

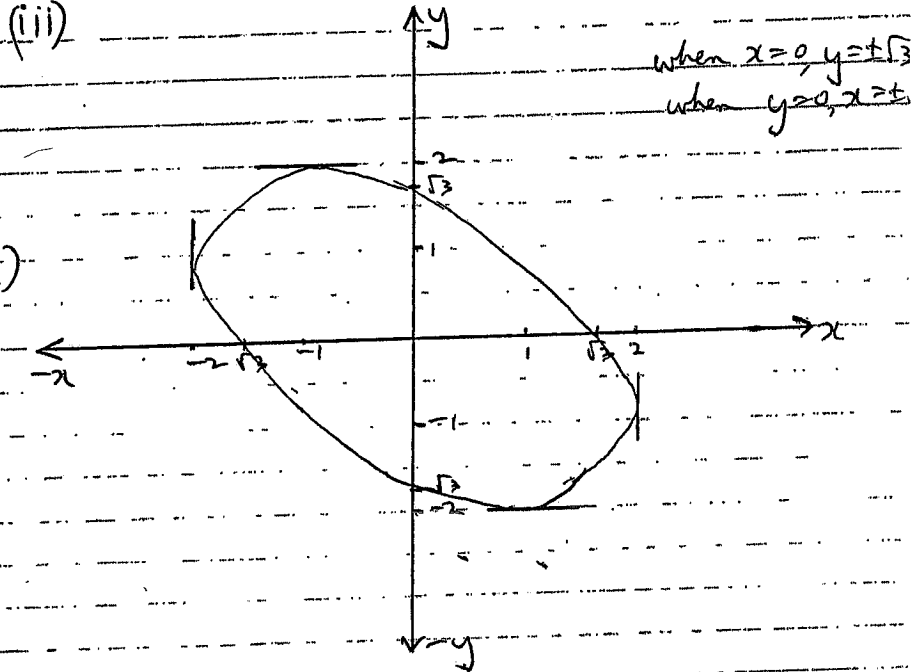
$$y = -1, x = 2$$

(2)

$\therefore y'$ undefined at $(-2, 1), (2, -1)$

(iii)

(2)



when $x = 0, y = \pm \sqrt{3}$

when $y = 0, x = \pm \sqrt{3}$

Q5

$$\begin{aligned}
 \text{Q) (i)} \quad I_n &= \int_0^{\frac{\pi}{2}} x^n \sin x \, dx, \quad n \geq 0 \\
 &= \int_0^{\frac{\pi}{2}} x^n \frac{d}{dx}(-\cos x) \, dx \\
 &= -x^n \cos x \Big|_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} \cos x \cdot n x^{n-1} \, dx \\
 &= n \int_0^{\frac{\pi}{2}} \cos x \cdot x^{n-1} \, dx \\
 &= n \int_0^{\frac{\pi}{2}} \frac{d}{dx}(\sin x) \cdot x^{n-1} \, dx \\
 &= n \left\{ x^{n-1} \sin x \Big|_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \sin x \cdot (n-1) x^{n-2} \, dx \right\} \\
 \therefore I_n &= n \left(\frac{\pi}{2}\right)^{n-1} \cdot 1 - 0 - n(n-1) \int_0^{\frac{\pi}{2}} x^{n-2} \sin x \, dx \quad (6)
 \end{aligned}$$

$$\therefore I_n + n(n-1) I_{n-2} = n \left(\frac{\pi}{2}\right)^{n-1} \quad \checkmark$$

$$\begin{aligned}
 \text{(5) (ii)} \quad I_1 &= \int_0^{\frac{\pi}{2}} x \sin x \, dx \\
 &= \int_0^{\frac{\pi}{2}} x \cdot \frac{d}{dx}(-\cos x) \, dx \\
 &= -x \cos x \Big|_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} \cos x \cdot 1 \, dx \\
 &= 0 + [\sin x]_0^{\frac{\pi}{2}} \\
 &= 0 + 1 - 0 \\
 &= 1
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad \text{let } n=3 \text{ in } \# \\
 \therefore I_3 + 3 \cdot 2 I_1 &= 3 \left(\frac{\pi}{2}\right)^2 \\
 \therefore \int_0^{\frac{\pi}{2}} x^3 \sin x \, dx = I_3 &= \frac{3\pi^2}{4} - 3 \cdot 2 \\
 &= \frac{3\pi^2}{4} - 6
 \end{aligned}$$

$$\int_0^1 \frac{x}{\sqrt{1-x^2}} \, dx$$

$$\begin{aligned}
 \text{let } u &= 1-x^2 & x=0, u=1 \\
 \therefore \frac{du}{dx} &= -2x & x=1, u=0 \\
 x \, dx &= -\frac{1}{2} du
 \end{aligned}$$

$$\begin{aligned}
 \therefore I &= \int_1^0 -\frac{1}{2} u^{-\frac{1}{2}} \, du \\
 &= \left[-\frac{1}{2} \cdot 2 u^{\frac{1}{2}} \right]_1^0 \\
 &= \left[-u^{\frac{1}{2}} \right]_1^0 \\
 &= 1
 \end{aligned}$$

$$\therefore \int_0^1 \cos^{-1} x \, dx = 0 - 0 + 1 = 1$$

$$\begin{aligned}
 \therefore V &= 2\pi \int_0^1 x \cos^{-1} x + \cos^{-1} x \, dx \\
 &= 2\pi \left(\frac{\pi}{8} + 1 \right) v^3
 \end{aligned}$$

Q4

$$\begin{aligned}
 \text{Q) (i)} \quad &1 + 10 + 10^2 + \dots + 10^n \\
 \text{G.P. } &a=1, r=10, N=n+1
 \end{aligned}$$

$$\begin{aligned}
 S_n &= a \frac{(r^N - 1)}{r - 1} \\
 &= \frac{10^{n+1} - 1}{10 - 1}
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad &1 \times 9^2 + 11 \times 9^2 + 111 \times 9^2 + \dots + \underbrace{(111\dots 1)}_{n \text{ ones}} \times 9^2 \\
 &= 10^{n+1} - 9^n
 \end{aligned}$$

b) at $t=0$, $x=0$, $v=0$.
 $\ddot{x}=1$ for $1 \leq t \leq 3$

$$\frac{dv}{dt} = 1, \quad 1 \leq t \leq 3$$

$$v = t + C$$

*

$$\begin{aligned} * \int_0^5 x \, dt &= v(5) - v(0) \\ &= v(5) \end{aligned}$$

$$\text{Now, } \int_0^1 \ddot{x} \, dt = \int_0^1 \frac{dv}{dt} \, dt \quad \therefore \text{velocity after 5s} \\ = v(t) \Big|_0^1 = 4 - \frac{1}{2} = 3\frac{1}{2}$$

$$= v(1) - v(0)$$

$$= v(1) \text{ since } v(0)=0$$

$\therefore$ vel. after 1 sec = $3\frac{1}{2}$ (area under graph)

when $t=1$, $v=3\frac{1}{2}$

Sub into *

$$\Rightarrow 3\frac{1}{2} = 1 + C$$

$$\therefore C = \frac{1}{2}$$

$$v = t + \frac{1}{2}, \quad 1 \leq t \leq 3$$

(ii) when $t=3$, $v=3\frac{1}{2}$, from (1)

Now, for $3 \leq t \leq 5$, $\ddot{x} = -t + 4$

$$\therefore v = -\frac{t^2}{2} + 4t + K$$

$$\text{when } t=3, v=3\frac{1}{2}$$

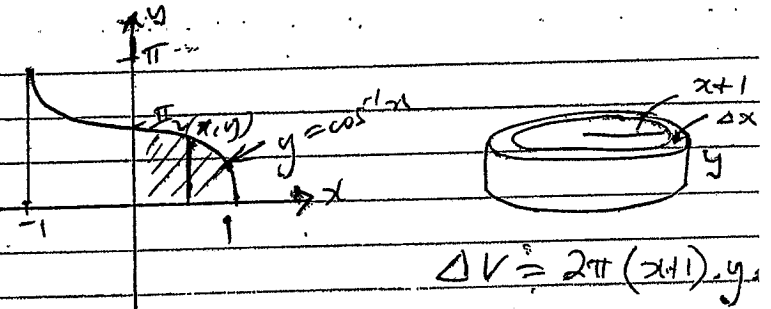
$$3\frac{1}{2} = -\frac{9}{2} + 12 + K$$

$$\therefore K = 8 - 12 = -4$$

$$\therefore v = -\frac{t^2}{2} + 4t - 4$$

when $t=5$,

c)



$$\Delta V = 2\pi(x+1)y \Delta x$$

$$\begin{aligned} V &= \lim_{\Delta x \rightarrow 0} \sum_{x=0}^1 2\pi(x+1)y \Delta x \\ &= 2\pi \int_0^1 (x+1) \cos^{-1} x \, dx \end{aligned}$$

$$\begin{aligned} \text{Now, } \int_0^1 x \cos^{-1} x \, dx &= \int_0^1 \cos^{-1} x \cdot \frac{d}{dx} \left(\frac{1}{2} x^2 \right) dx \\ &= \frac{1}{2} x^2 \cos^{-1} x \Big|_0^1 - \int_0^1 x \cdot \frac{-1}{\sqrt{1-x^2}} dx \end{aligned}$$

$$= \frac{1}{2} x^2 \cos^{-1} x \Big|_0^1 + \frac{1}{2} \int_0^1 \frac{x^2}{\sqrt{1-x^2}} dx$$

$$\int_0^1 \frac{x^2}{\sqrt{1-x^2}} dx = -\int_0^1 \frac{1-x^2}{\sqrt{1-x^2}} dx + \int_0^1 \frac{dx}{\sqrt{1-x^2}}$$

(6)

$$\text{Area} = \frac{1}{4} \pi$$

$$= -\int_0^1 \sqrt{1-x^2} \, dx + \left[\sin^{-1} x \right]_0^1$$

$$= -\frac{\pi}{4} + \sin^{-1} 1 - \sin^{-1} 0$$

$$= -\frac{\pi}{4} + \frac{\pi}{2} = \frac{\pi}{4}$$

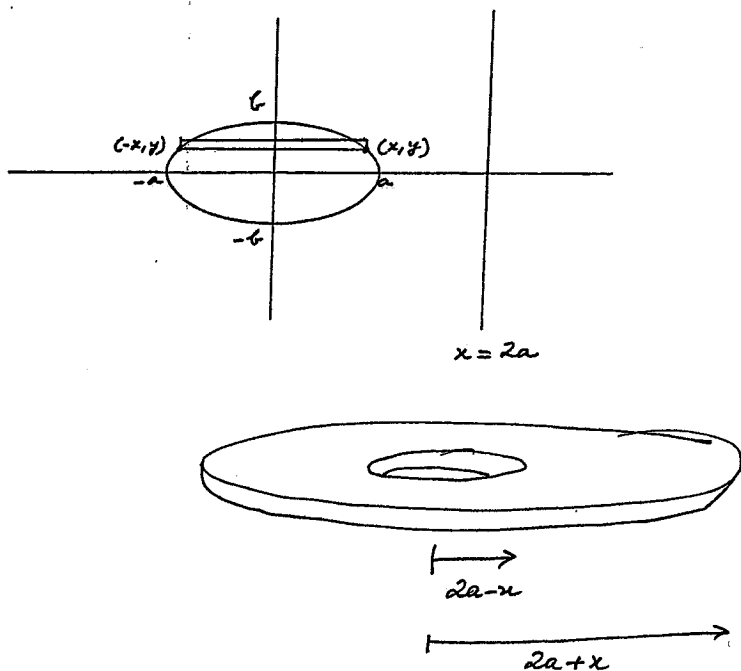
$$\begin{aligned} \therefore \int_0^1 x \cos^{-1} x \, dx &= 0 - 0 + \frac{1}{2} \times \frac{\pi}{4} \\ &= \frac{\pi}{8} \end{aligned}$$

$$\begin{aligned} \text{Also, } \int_0^1 \cos^{-1} x \, dx &= \int_0^1 \cos^{-1} x \cdot \frac{d}{dx} x \, dx \\ &= x \cos^{-1} x \Big|_0^1 + \int_0^1 \frac{x}{\sqrt{1-x^2}} dx \end{aligned}$$

OR *

QUESTION 6:

(a)



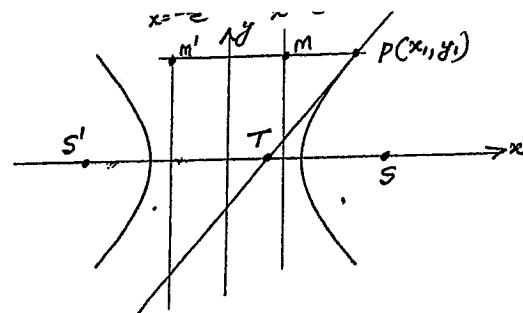
Volume of slice is

$$\begin{aligned} \delta V &= \pi(2a+x)^2 \delta y - \pi(2a-x)^2 \delta y \\ &= \pi[(2a+x+2a-x)(2a+x-2a+x)] \delta y \\ &= \pi(4a)(2x) \delta y \\ &= 8\pi ax \delta y \end{aligned}$$

$\therefore$ Volume of solid is

$$\begin{aligned} V &= \lim_{\delta y \rightarrow 0} \sum_{y=-b}^b 8\pi ax \delta y \\ &= 8\pi a \int_{-b}^b x dy \text{ where } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \\ &= 8\pi a \int_{-b}^b \frac{a}{b} (b^2 - y^2)^{\frac{1}{2}} dy \quad \frac{x^2}{a^2} = 1 - \frac{y^2}{b^2} \\ &= 8\pi a \times \frac{a}{b} \times \pi b^2 \text{ (area of } \end{aligned}$$

(b)



$$(i) \quad \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

Differentiating with respect to x

$$\Rightarrow \frac{2x}{a^2} - \frac{2y}{b^2} \frac{dy}{dx} = 0$$

$$\text{ie } \frac{dy}{dx} = + \frac{b^2 x}{a^2 y}$$

$$\text{at } P(x_1, y_1) \quad \frac{dy}{dx} = + \frac{b^2 x_1}{a^2 y_1}$$

$\therefore$ Tangent at $P(x_1, y_1)$ is

$$y - y_1 = + \frac{b^2 x_1}{a^2 y_1} (x - x_1)$$

$$a^2 y y_1 - a^2 y_1^2 = + b^2 x x_1 - b^2 x_1^2$$

$$\text{ie } b^2 x x_1 - a^2 y y_1 = b^2 x_1^2 - a^2 y_1^2$$

$$\text{Q.E.D.} \Rightarrow \frac{x x_1}{a^2} - \frac{y y_1}{b^2} = \frac{x_1^2}{a^2} - \frac{y_1^2}{b^2}$$

$$= 1 \text{ since } (x_1, y_1) \text{ lies on } \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

(ii) at T , $y = 0$

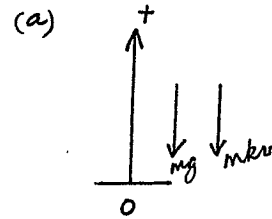
$$\therefore \frac{x x_1}{a^2} = 1$$

$$\text{ie } x = \frac{a^2}{x_1}$$

$$\therefore T \equiv \left(\frac{a^2}{x_1}, 0 \right)$$

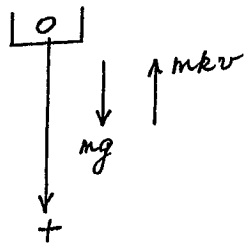
$$\begin{aligned}
 \text{(iii)} \quad \frac{PS}{PS'} &= \frac{ePM}{ePM'} \\
 &= \frac{PM}{PM'} \\
 &= \frac{x_1 - \frac{a}{e}}{x_1 + \frac{a}{e}} \\
 &= \frac{x_1 e - a}{x_1 e + a}
 \end{aligned}
 \qquad
 \begin{aligned}
 \frac{TS}{TS'} &= \frac{ae - \frac{a^2}{x_1}}{\frac{a^2}{x_1} + ae} \\
 &= \frac{ae x_1 - a^2}{a^2 + ae x_1} \\
 &= \frac{a(x_1 e - a)}{a(a + x_1 e)} \\
 &= \frac{PS}{PS'}
 \end{aligned}$$

QUESTION 7:



$$\begin{aligned}
 R &= m\ddot{x} \\
 \Rightarrow m\ddot{x} &= -mg - mkv \\
 \text{ie } \ddot{x} &= -g - kv \\
 \therefore v \frac{dv}{dx} &= -g - kv \\
 \frac{dv}{dx} &= \frac{-g - kv}{v} \\
 \therefore \frac{dx}{dv} &= -\frac{v}{g + kv}
 \end{aligned}$$

$$\begin{aligned}
 \therefore x_{\max} &= -\int_v^0 \frac{v}{g + kv} dv \\
 &= -\int_0^v \left(\frac{1}{k} \cdot \frac{g + kv}{g + kv} - \frac{g}{k} \cdot \frac{1}{g + kv} \right) dv \\
 &= -\int_0^v \left[\left(\frac{1}{k} \right) \cdot 1 - \frac{g}{k} \cdot \frac{k}{g + kv} \right] dv \\
 &= -\left[\frac{v}{k} - \frac{g}{k} \ln(g + kv) \right]_v^0 \\
 &= \frac{g}{k} \ln g + \frac{V}{k} - \frac{g}{k} \ln(g + kV) \\
 &= \frac{V}{k} + \frac{g}{k} \ln\left(\frac{g}{g + kV}\right)
 \end{aligned}$$



$$\begin{aligned}
 \text{(i)} \quad R &= m\ddot{x} \\
 \Rightarrow mg - mkv &= m\ddot{x} \\
 g - kv &= \ddot{x} \quad \text{--- (1)}
 \end{aligned}$$

Terminal velocity at $\ddot{x} = 0$

$$\begin{aligned}
 \text{ie } g - kv &= 0 \\
 \therefore v &= \frac{g}{k}
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad \textcircled{1} \Rightarrow \frac{dv}{dt} &= g - kv \\
 \therefore \frac{dt}{dv} &= \frac{1}{g - kv} \\
 t &= \int_0^{\frac{g}{k}} \frac{1}{g - kv} dv \\
 &= -\frac{1}{k} \ln(g - kv) \Big|_0^{\frac{g}{k}} \\
 &= -\frac{1}{k} \ln\left(g - \frac{g}{k} \cdot \frac{g}{k}\right) + \frac{1}{k} \ln g \\
 &= \frac{1}{k} \ln\left(\frac{g}{g - \frac{g^2}{k^2}}\right)
 \end{aligned}$$

(b)

$$A+B+C = 180$$

$$\therefore A+B = 180-C$$

$$\therefore \tan(A+B) = \tan(180-C)$$

$$= -\tan C$$

$$\Rightarrow \frac{\tan A + \tan B}{1 - \tan A \tan B} = -\tan C$$

$$\therefore \tan A + \tan B = -\tan C + \tan A \tan B \tan C$$

$$\text{ie } \tan A + \tan B + \tan C = \tan A \tan B \tan C$$

$$(c) \quad (i) \quad \tan 4x = \frac{2 \tan 2x}{1 - \tan^2 2x}$$

$$= \frac{2 \cdot \left(\frac{2 \tan x}{1 - \tan^2 x} \right)}{1 - \left(\frac{2 \tan x}{1 - \tan^2 x} \right)^2}$$

$$= \frac{2 \cdot \frac{2t}{1-t^2}}{1 - \left(\frac{2t}{1-t^2} \right)^2} \times \frac{(1-t^2)^2}{(1-t^2)^2}$$

$$= \frac{4t(1-t^2)}{(1-t^2)^2 - 4t^2}$$

$$= \frac{4t(1-t^2)}{t^4 - 6t^2 + 1}$$

$$(ii) \quad \tan x \tan 4x = 1$$

$$\Rightarrow t \cdot \frac{4t(1-t^2)}{t^4 - 6t^2 + 1} = 1$$

$$4t^2(1-t^2) = t^4 - 6t^2 + 1$$

$$4t^2 - 4t^4 = t^4 - 6t^2 + 1$$

$$(iii) \quad x = 18^\circ : \tan 18^\circ \tan 72^\circ$$

$$= \cot 72^\circ \cdot \tan 72^\circ$$

$$= 1$$

$$x = 54^\circ : \tan 54^\circ \tan 216^\circ$$

$$= \cot 36^\circ \cdot \tan 36^\circ$$

$$= 1$$

$$\therefore \text{Both } x = 18^\circ \text{ and } x = 54^\circ \text{ satisfy}$$

$$\tan x \tan 4x = 1$$

$$(iv) \quad 5t^4 - 10t^2 + 1 = 0 \text{ has solutions}$$

$$t = \tan 18^\circ, \tan 54^\circ$$

$$\Rightarrow t^2 = \frac{10 \pm \sqrt{100 - 20}}{10}$$

$$= \frac{10 \pm 4\sqrt{5}}{10}$$

$$= \frac{5 \pm 2\sqrt{5}}{5}$$

$$= \frac{5 - 2\sqrt{5}}{5}, \frac{5 + 2\sqrt{5}}{5}$$

$$\text{ie } t = \pm \sqrt{\frac{5 - 2\sqrt{5}}{5}}, \pm \sqrt{\frac{5 + 2\sqrt{5}}{5}}$$

Since $\tan 18^\circ$ and $\tan 54^\circ$ are both positive with $\tan 54^\circ > \tan 18^\circ$ Then

$$\tan 54^\circ = \left(\frac{5 + 2\sqrt{5}}{5} \right)^{\frac{1}{2}}$$

QUESTION 8:

(a) (i) $\underbrace{1 + 10 + 10^2 + \dots + 10^n}_{\text{Geometric series } a=1, r=10, n=n+1}$

$$S = 1 \left[\frac{10^{n+1} - 1}{10 - 1} \right]$$

$$= \frac{10^{n+1} - 1}{9} \quad \text{--- (1)}$$

(ii) Let $S(n)$ be the assertion that
 $1 \times 9^2 + 11 \times 9^2 + \dots + \underbrace{(11 \dots 1)}_{n \text{ ones}} 9^2 = 10^{n+1} - 9n - 10$

Test $S(1)$: LHS = 1×9^2
 $= 81$
 RHS = $10^2 - 9 - 10$
 $= 81$

$\therefore S(1)$ is true

Assume $S(k)$ is true for some integer $n=k$
 i.e. $1 \times 9^2 + 11 \times 9^2 + \dots + \underbrace{(11 \dots 1)}_{k \text{ ones}} 9^2 = 10^{k+1} - 9k - 10$

(Now $1 \times 9^2 + 11 \times 9^2 + \dots + \underbrace{(11 \dots 1)}_{k \text{ ones}} 9^2 + \underbrace{(11 \dots 1)}_{(k+1) \text{ ones}} 9^2$
 $= 10^{k+1} - 9k - 10 + 9^2 \left[1 + 10 + 10^2 + \dots + 10^k \right]$
 $= 10^{k+1} - 9k - 10 + 9^2 \cdot \left(\frac{10^{k+1} - 1}{9} \right)$ from (1)
 $= 10^{k+1} - 9k - 10 + 9 \cdot 10^{k+1} - 9$
 $= 10^{k+1} [1 + 9] - 9(k+1) - 10$
 $= 10^{k+2} - 9(k+1) - 10$
 $= 10^{n+1} - 9n - 10$ where $n = k+1$

Hence if $S(n)$ is true for $n=k$ then it is also true for $n=k+1$.

But true for $n=1 \Rightarrow$ true for $n=2$

and then by the principle of mathematical induction $S(n)$ is true for all $n \geq 1$

(b) (i) $z^5 = 1 \quad z \neq 1$

$\Rightarrow z^5 - 1 = 0$

$(z-1)(z^4 + z^3 + z^2 + z + 1) = 0$

$z \neq 1 \Rightarrow z^4 + z^3 + z^2 + z + 1 = 0$

$\Rightarrow z^4 + z^3 + 1 + z^{-1} + z^{-2} = 0$ --- (1)

(ii) $z^5 = 1$
 $= 1 \cos(0 + 2k\pi)$

$\therefore z = \cos \frac{2k\pi}{5} \quad k=0, 1, 2, 3, 4$

but $z \neq 1 \Rightarrow z = \cos \frac{2k\pi}{5} \quad k=1, 2, 3, 4$

Then $z + z^{-1} = \cos \frac{2k\pi}{5} + i \sin \frac{2k\pi}{5} + \cos \left(-\frac{2k\pi}{5} \right) + i \sin \left(-\frac{2k\pi}{5} \right)$
 $= 2 \cos \frac{2k\pi}{5} \quad k=1, 2, 3, 4$

(iii) If $x = z + z^{-1}$

$x^2 = z^2 + 2 + z^{-2} \Rightarrow z^2 + z^{-2} = x^2 - 2$

Then (1) $\Rightarrow (z^2 + z^{-2}) + (z + z^{-1}) + 1 = 0$

$\Rightarrow x^2 - 2 + x + 1 = 0$

$x^2 + x - 1 = 0$ --- (2)

$$\therefore \therefore \text{now } z + z^{-1} = 2 \cos \frac{2k\pi}{5} \quad k=1, 2, 3, 4$$

$$k=1 \Rightarrow z + z^{-1} = 2 \cos \frac{2\pi}{5}$$

$$k=2 \Rightarrow z + z^{-1} = 2 \cos \frac{4\pi}{5} = -2 \cos \frac{\pi}{5}$$

$$k=3 \Rightarrow z + z^{-1} = 2 \cos \frac{6\pi}{5} = -2 \cos \frac{\pi}{5}$$

$$k=4 \Rightarrow z + z^{-1} = 2 \cos \frac{8\pi}{5} = 2 \cos \frac{2\pi}{5}$$

Hence the solutions of (2) are

$$x = 2 \cos \frac{2\pi}{5} \text{ and } x = -2 \cos \frac{\pi}{5}$$

○ (

Product of roots

$$\Rightarrow 2 \cos \frac{2\pi}{5} \times -2 \cos \frac{\pi}{5} = \frac{c}{a} = -1$$

$$\therefore -4 \cos \frac{\pi}{5} \cos \frac{2\pi}{5} = -1$$

$$\therefore \cos \frac{\pi}{5} \cos \frac{2\pi}{5} = \frac{1}{4}$$

○ (